



HyperPoint: Multimodal 3D foundation model in hyperbolic space

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ABSTRACT

Self-supervised learning has made significant progress in Natural Language Processing and Computer Vision. Nevertheless, it encounters significant obstacles in the 3D domain, primarily due to the scarcity of available data and the considerable challenge of effectively capturing hierarchical structures. Current methods in Euclidean space suffer from feature distortion and fail to model the semantic hierarchies inherent in cross-modal data. These challenges motivate us to adopt hyperbolic space, which excels at capturing multi-scale relationships and preserving the geometric structure of complex data. In this paper, we propose HyperPoint, the first multimodal 3D foundational model in hyperbolic space. By projecting cross-modal features such as 3D point cloud, 2D images, and text into hyperbolic space, we leverage its tree-like properties to encode semantic hierarchies with minimal distortion. Our method integrates generative and contrastive learning while leveraging multi-loss optimization to enhance feature diversity and consistency. *HyperPoint* achieves a new state-of-the-art in 3D representation learning, e.g., **96.1 %** accuracy on ScanObjectNN, **94.1 %** accuracy on 10w10s on ModelNet40. Our code is available at: <https://github.com/Issac-Sun/HyperPoint>.

1. Introduction

Point cloud is a fundamental and widely used method for representing 3D objects. It captures abundant geometric information, enabling a comprehensive understanding of the spatial structure of objects. This characteristic makes point cloud indispensable in various application fields such as autonomous driving and robotics.

In recent years, self-supervised representation learning has advanced significantly in Natural Language Processing (NLP) [1,2], Computer Vision (CV) [3], and cross-modal feature fusion domains [4,5]. However, in 3D Computer Vision, limited annotated data pose unique challenges [6–8], making it difficult to fully unlock the potential of 3D data. To address these challenges, researchers have explored leveraging data from other modalities to improve point cloud representation learning. Yet, the lack of effective feature extraction methods for point cloud constrains a better performance. Through in-depth analysis, we attribute three factors that hinder the further development of point cloud foundation models.

A. Geometric capacity

Recent work [9] shows that real-world datasets often exhibit natural hierarchies or inherent label structures visible as power-law and tree-like distributions. However, the volume of Euclidean space grows only polynomially with the radius, so the capacity is bounded when the embedding dimension is limited, a well-known manifestation of the “curse of dimensionality”.

B. Distortion lower bound

Packing an exponentially growing set of neighbours into a polynomially growing volume crowds the feature space and makes node embeddings hard to separate. This effect is called distortion, and it prevents the Euclidean distance between embeddings from reflecting the hop count between nodes [10]. Consequently, distortion makes it hard to infer object labels from Euclidean distances, forcing the model to treat labels as permutation invariant. Such compression inevitably discards the rich semantic information contained in label hierarchies, limiting the model's ability to recover or infer the hierarchy from the label space.

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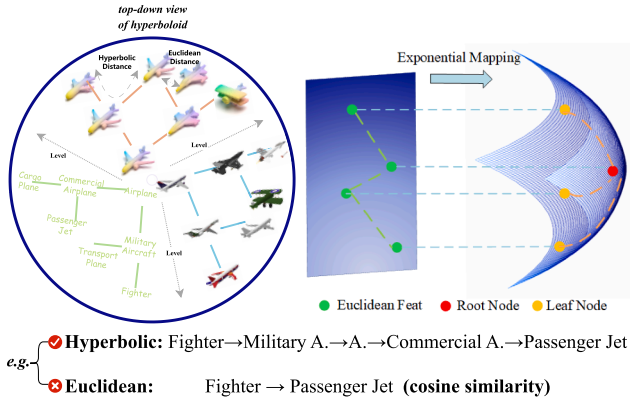


Fig. 1. Top-Down View of a Hyperboloid Structure (left). The schematic diagram of the exponential mapping from Euclidean space to hyperbolic space (right). The hierarchical tree embeds three modals under the category of ‘Airplane.’ Lower-level leaf nodes contain more fine-grained semantic information. When calculating inter-class distances, point cloud features from nodes at the same hierarchical level should traverse through the root node. Specifically, the concept ‘Fighter Aircraft’ is semantically closer to the general concept ‘Airplane’ than to another specific concept ‘Passenger Jet’. This manifold effectively captures the semantic hierarchy of the data. A. represents Airplane / Aircraft for short.

C. Intrinsic structure of 3D point clouds

Euclidean space is suboptimal for representing hierarchies because it is flat [11]. It cannot capture the distances between elements accurately, and therefore neglects the semantic similarity of visual parts. Point clouds can be viewed as hierarchical part-whole compositions. Recognizing an instance at coarse granularity (e.g., an aeroplane) requires understanding its parts at finer granularity (wings, nose, fuselage). Aeroplanes and fighters look similar at coarse level but differ markedly at fine level. If the visual hierarchy is not fully encoded in the global feature, the 3D understanding of a pretrained foundation model will remain limited.

We argue that these limitations stem from ignoring the hierarchical structure inherent in point cloud data, leading to a critical challenge: how can we maximize the utilization of hierarchical structures across modalities to effectively aggregate features?

We identify hyperbolic space as an ideal solution to this problem. Hyperbolic volume grows exponentially, so hyperbolic embeddings offer far larger geometric capacity than Euclidean embeddings of the same dimension. Unlike Euclidean space, hyperbolic space has constant negative curvature, so it can embed the semantic tree of a 3D point cloud with minimal distortion. Most importantly, Fig. 1 shows that hyperbolic space can measure the semantic structure of point clouds without introducing structural error.

Further, we wonder that how to build a cross-modal 3D foundation model in hyperbolic space. In Appendix A we show the limitations of existing contrastive and generative learners as ShapeNet [12] scales from 1% to 100%. With contrastive loss alone, the model already overfits at 25% of the data and learns spurious shortcuts [3]. This indicates that contrastive learning is sensitive to negative-sample density. When data are scarce, the signal-noise ratio drops sharply. Conversely, when pre-training data exceed 75%, MAE-style gains saturate while contrastive learning still yields large improvements. This shows that generative methods lack further scaling ability [6] and cannot build cross-part semantic links, limiting fine-grained classification.

In this work, we propose HyperPoint, the first multimodal 3D foundation model in hyperbolic space. We project cross-modal features, including 3D point clouds, 2D images, and text, into hyperbolic space to leverage its tree-like properties for encoding semantic hierarchies with minimal distortion. Our method integrates generative and contrastive learning, using a multi-loss optimization approach that balances feature diversity and semantic consistency. Specifically, we apply a hybrid

masking strategy for the point cloud encoder and introduce a novel entailment loss to enforce logical relationships between different modalities in hyperbolic space. Through extensive experiments, we achieve a total accuracy improvement of +4.15% and +18.33% over current cross-modal contrastive learning [13] and single-modal MAE methods [14] in downstream tasks. In four few-shot downstream tasks, HyperPoint continues to lead, achieving improvements of +1.6% and +4.6% respectively. The main contributions of our method are summarized as follows:

- We propose HyperPoint, the first multimodal 3D foundation model in hyperbolic space. Our method not only unifies generative and contrastive learning but also extends cross-modal contrastive learning to hyperbolic space.
- We propose a comprehensive multi-loss optimization method to tackle the challenges of hierarchical and multi-modal complexities in point cloud representation learning. This method integrates generative, contrastive, and structural learning objectives into a unified design, ensuring a balance between feature diversity and semantic consistency.
- We perform thorough qualitative analyses with HyperPoint to demonstrate its potential for capturing hierarchical structures and preserving semantic consistency in cross-modal data. Through extensive experiments, we demonstrate that the hyperbolic counterpart outperforms the Euclidean setting.

2. Related works

Contrastive Learning. Contrastive representation learning emerged as a predominant paradigm in self-supervised learning, leveraging the principle of alignment and separation of sample representations. Traditionally, contrastive learning maximized agreement between augmented views of the positive samples while ensuring dissimilarity with negative samples. Methods such as SimCLR [15] and CLIP [4] extended this paradigm to single-modal and cross-modal contexts respectively. Within the 3D domain, methods like PointContrast [16] adapted these principles to point cloud by introducing geometric augmentations, while CrossPoint [5] integrated inter-modal and intra-modal learning. Unlike previous methods, HyperPoint extended contrastive learning into hyperbolic space and introduced a novel hyperbolic contrastive loss.

Generative Learning. Generative masked representation learning built on the success of masked autoencoders (MAE) [3] in NLP and 2D vision. By reconstructing masked input data, these methods enforced models to capture structured representations. In 3D vision, PointMAE [14] pioneered the adaptation of MAE for point cloud, utilizing masking strategies to learn robust global features. Subsequent works, such as PointM2AE [17], introduced multi-scale hierarchies and sophisticated masking mechanisms to enhance performance. Cross-modal extensions like I2P-MAE [18] and ACT [6] leveraged 2D and language data for guidance, further improving pre-training efficacy. Despite these advancements, challenges remained in integrating global and local feature learning, as generative models tended to focus on localized attention patterns. Diffusion-based representation learning models are also beginning to emerge. Pointdiff [19] performed iterative denoising on the noisy point cloud to acquire a comprehensive understanding of the original one. Hybrid methods like RECON [13] addressed this by combining generative and contrastive learning, achieving superior generalization on downstream tasks. Leveraging the superior properties of hyperbolic geometry, our method outperformed it in all downstream tasks.

Hyperbolic Embedding. Hyperbolic geometry gained significant traction in representation learning due to its ability to model hierarchical data efficiently [10,11]. Unlike Euclidean space, hyperbolic space allowed for the representation of tree-like structures with minimal distortion, making them ideal for capturing semantic hierarchies in multimodal and 3D data. Early applications in NLP demonstrated the

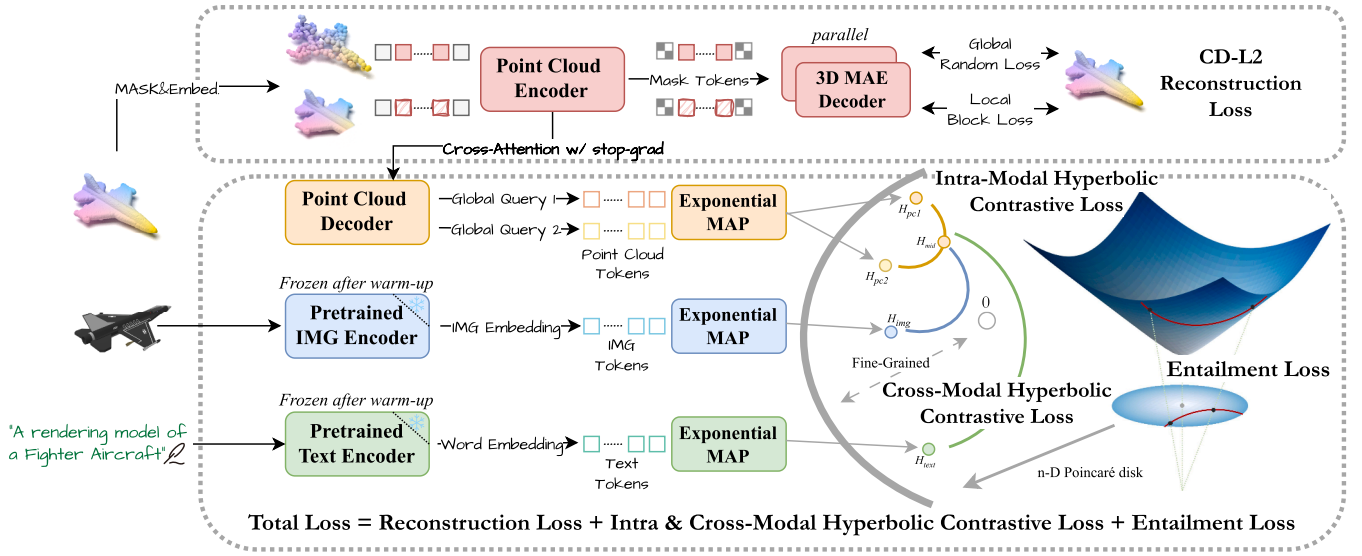


Fig. 2. Pipeline of our HyperPoint framework. The input modalities: point cloud, image, and text are encoded into tokens through respective encoders. For the 3D token embeddings, a hybrid masking strategy is applied, combining local block masking and global random masking for reconstruction. Noted that intermediate embeddings are fed into the 3D decoder via cross-attention with stop-gradient operation. These embeddings are then utilized for subsequent hyperbolic projection and cross-modal contrastive learning.

potential of hyperbolic embeddings to represent linguistic hierarchies, while more recent methods like MERU [20] extended this to vision-language tasks. In 3D vision, HyCoRe [21] captured 3D part-whole hierarchies in supervised learning. HyperIPC [22] introduced hyperbolic contrastive learning to encode both intra-modal and cross-modal hierarchies, leveraging the intrinsic properties of hyperbolic space to enhance representation compactness and generalization. HypStructure [9] embedded label hierarchies into hyperbolic space with minimal distortion, boosting classification and OOD detection. Hi-Mapper [11] modeled visual hierarchies with Gaussian mixtures and learned them in hyperbolic space via a novel hierarchical contrastive loss. HHCH [23] introduced hierarchical hyperbolic contrastive hashing, embedding continuous codes into the Poincaré ball with a novel hyperbolic K-Means and hierarchical contrastive learning to exploit visual hierarchies for superior self-supervised retrieval. HyperCL [10] fused contrastive learning with hyperbolic graph and achieved superior recommendation accuracy. More details refer to Appendix C. By mapping point cloud features into hyperbolic space, our method surpasses Euclidean-based methods.

3. Method

Our model's pipeline is illustrated in Fig. 2. In this section, we detail the process of hyperbolic contrastive learning, describe how generative and contrastive methods work in synergy, and outline the loss function. Specifically, in Appendix B, we present essential computational formulas in hyperbolic space along with foundational knowledge. Section 3.1 introduces a novel approach to contrastive learning within hyperbolic space, enhancing the representation of hierarchical structures in point clouds. Section 3.2 utilizes generative MPM (Mask Point Modeling) task to improve the robustness of point cloud feature learning. Section 3.3 implements an entailment loss function to enforce a logical relationship between different modalities in the hyperbolic space.

3.1. Hyperbolic geometric embedding

Hyperbolic geometry is a type of non-Euclidean geometry characterized by constant negative curvature. There are multiple hyperbolic models with different definitions and metrics. Since the Poincaré disk

model maintains numerical stability better during the gradient-based learning process, we adopt it in our work, defined as $(\mathbb{V}_C^n, g^V)$.

A. Space definition and basic operations

The manifold $\mathbb{V}_C^n = \{x \in \mathbb{R}^n : C\|x\|^2 < 1, C \geq 0\}$ is equipped with a Riemannian metric $g^V = \lambda_C^x g^E$, where g^E represents the metric tensor, and $\lambda_C^x = \frac{2}{1-C\|x\|^2}$ is a conformal factor dependent on the curvature C and the position of the calculated point x . For clarity, $\|\cdot\|$ is defined as the norm, and $\cdot$ as the inner product. Given a pair of vectors $u, v \in \mathbb{V}_C^n$, the addition is defined as:

$$u \oplus_C v = \frac{(1 + 2Cu \cdot v + C\|v\|^2)u + (1 - C\|u\|^2)v}{1 + 2Cu \cdot v + C^2\|u\|^2\|v\|^2}. \quad (1)$$

In hyperbolic space, the distance between u and v is defined as:

$$D_{\text{hyp}}(u, v) = \frac{2}{\sqrt{C}} \operatorname{arctanh} \left(\sqrt{C} \| -u \oplus_C v \| \right), \quad (2)$$

when C approaches zero, the formula reduces to the traditional Euclidean metric.

In hyperbolic space, a geodesic is a generalization of the shortest path between two points or two planes. As the curvature decreases, the distance between two points increases, and the geodesic approaches the boundary. Due to the tree-like structure of hyperbolic space, the midpoint of a geodesic lies close to the origin and can be regarded as an abstract parent node. As illustrated in Fig. 3, points near the boundary represent more specific categories, while points closer to the origin represent more generalized instances. Given n hyperbolic node vectors $(M_1, M_2, \dots, M_n)$, their midpoint can be computed in the gyrovector space as:

$$M_{\text{mid}} = \frac{1}{2} \oplus_C \left(\frac{\sum_{i=1}^n \lambda_C^x M_i}{\sum_{i=1}^n (\lambda_C^x - 1)} \right). \quad (3)$$

B. From Euclidean features to the disk

To map embeddings into hyperbolic space, we define a mapping from Euclidean space to the Poincaré disk, referred as the exponential map. The hyperbolic manifold $\mathbb{V}_C^n$ at x has a first-order linear approximation in the tangent space $\mathcal{T}_x \mathbb{V}_C^n \cong \mathbb{R}^n$, where $\mathcal{T}_x \mathbb{V}_C^n = \{w \in \mathbb{R}^d : \langle w, x \rangle = 0\}$. The exponential map is defined as:

$$\exp_x^C(w) = x \oplus_C \left(\tanh \left(\sqrt{C} \frac{\lambda_C^x \|w\|}{2} \right) \frac{w}{\sqrt{C}\|w\|} \right). \quad (4)$$

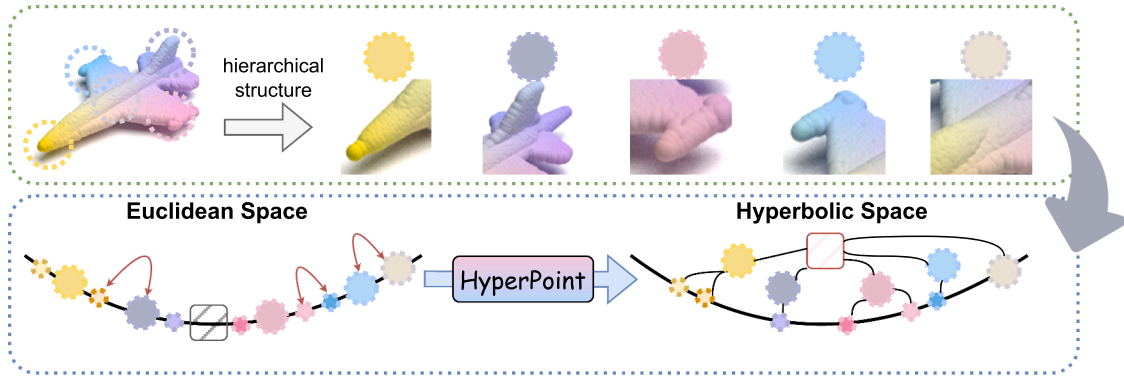


Fig. 3. Schematic of hyperbolic embedding. Euclidean embedding ignores the hierarchical structure of point clouds. Hyperbolic space fully encodes this visual hierarchy into global features, improving the 3D understanding of foundation models.

C. Midpoint and contrastive losses

Upon completing the aforementioned operations, our hyperbolic contrastive learning model can be divided into two distinct branches: intra-modal and cross-modal hyperbolic contrastive learning. These branches capture the semantic hierarchy of point clouds from different perspectives. Given the sample data $D = \{(P_i, I_i, T_i)\}_{i=1}^{|D|}$, where P_i, I_i, T_i denote 3D point cloud, 2D Image, Text. We enhance the capability of our point cloud encoder by applying rotation augmentation, generating two point clouds, P_{r1} and P_{r2} . A shared-weight point cloud decoder is employed to extract global features H_{pc1} and H_{pc2} . These latent features are then mapped to hyperbolic space using the exponential map mentioned in Eq. 4. Generally, given the representation pair (M_i, M_j) , we define the contrastive loss in hyperbolic space as follows:

$$l(M_i, M_j) = -\log \frac{\exp(-D_{\text{hyp}}(M_i, M_j)/\tau)}{\sum_{k=1, k \neq i}^N \exp(-D_{\text{hyp}}(M_i, M_k)/\tau)}, \quad (5)$$

where D_{hyp} is the distance calculated by hyperbolic space in Eq. 2, τ is the temperature coefficient, and N represents the number of samples in a batch. The loss is calculated by all the positive samples (i, j) and (j, i) .

$$\mathcal{L}(M_i, M_j) = \frac{1}{2N} \sum_{i=1}^N [l(M_i, M_j) + l(M_j, M_i)]. \quad (6)$$

We compute the midpoint using the previously mentioned in Eq. 3. HyperIPC [22] and HyperViT [24] noted that images and text exhibit more hierarchical semantic information in hyperbolic space. Therefore, we adopt a similar method to map 2D and text features into hyperbolic space, obtaining H_{img} and H_{text} . We leverage the hierarchical information from text and images to maximize the similarity between the midpoint H_{mid} and both H_{img} and H_{text} , thereby enhancing the discriminative capability of the point cloud model. The loss function for the hyperbolic contrastive learning module consists of three components.

$$\mathcal{L}_{\text{contrastive}} = \mathcal{L}(H_{\text{pcl}}, H_{\text{pc2}}) + \mathcal{L}(H_{\text{mid}}, H_{\text{img}}) + \mathcal{L}(H_{\text{mid}}, H_{\text{text}}). \quad (7)$$

3.2. Generative masked modeling

Following PointMAE [14], we process the input point cloud through three stages: patch generation, hybrid masking, and embedding. Given a point cloud $\mathbf{PC} \in \mathbb{R}^{N \times 3}$, we divide the input point cloud into irregular p point patches $\mathbf{P} \in \mathbb{R}^{p \times m \times 3}$ with m local points by Farthest Point Sampling (FPS) and the K-Nearest Neighbors (KNN) algorithm.

$$\begin{aligned} \mathbf{C} &= \text{FPS}(\mathbf{X}), & \mathbf{C} &\in \mathbb{R}^{n \times 3}, \\ \mathbf{P} &= \text{KNN}(\mathbf{X}, \mathbf{C}), & \mathbf{P} &\in \mathbb{R}^{n \times k \times 3}. \end{aligned} \quad (8)$$

Notably, to demonstrate the effectiveness of our hyperbolic contrastive learning, we adopt a novel hybrid masking strategy. Specifically, we apply two parallel masking strategies to the input point cloud: **local** block masking and **global** random masking. Generate $\mathbf{P}_m^l \in \mathbb{R}^{rp \times m \times 3}$

and $\mathbf{P}_m^g \in \mathbb{R}^{rp \times m \times 3}$ with masking ratio r as masked patches respectively. For embedding, $\mathbf{P}_v$ is processed using a mini-PointNet [25] to produce encoded tokens $\mathbf{E}_v$, which can be represented as:

$$\mathbf{E}_v = \text{PointNet}(\mathbf{P}_v), \quad \mathbf{E}_v \in \mathbb{R}^{n \times D}, \quad (9)$$

where D denotes the hidden dimension of the network.

Recent studies [13,26,27] show that PointMAE [14] lacks explicit spatial locality and cannot aggregate context across patches. Inspired by EdgeConv [28], we introduce the Geometric Calibration Module (GCM). GCM performs patch-level graph convolution on a dynamically built k-nearest neighbour graph, so each patch can gather features from the whole space without being limited to its own window. Given a patch token E_v^i , we define the edge feature between patch i and patch j as:

$$e_{ij} = \phi([E_v^i - E_v^j; E_v^i]), \quad (10)$$

where $\phi: \mathbb{R}^{2C} \rightarrow \mathbb{R}^C$ is a learnable MLP. We then aggregate the edge features with max pooling and project back to dimension C :

$$E_v^i = \xi\left(\max_{j:(i,j) \in \mathcal{E}} e_{ij}\right), \quad (11)$$

where $\xi: \mathbb{R}^C \rightarrow \mathbb{R}^C$ is another MLP. In this way, GCM explicitly combines the global variation $E_v^j - E_v^i$ with the local structure E_v^i . By gathering information across multiple scales, it learns the overall geometry of the scene and improves fine-grained 3D perception.

We then employ two distinct decoders, Dec^g and Dec^l , both structured identically. In the local branch, we first concatenate the encoder output E_v^l with randomly initialized learnable mask tokens $\mathbf{E}_m^l$ and direct this composite input into Dec^l . Similar processes are undertaken for the global branch. After passing through two independent decoders and linear heads, we yield $\mathbf{R}_m^l \in \mathbb{R}^{rp \times m \times 3}$ and $\mathbf{R}_m^g \in \mathbb{R}^{rp \times m \times 3}$, calculate the reconstruction loss using the l_2 Chamfer Distance (CD), defined as:

$$\mathcal{L}_{\text{reconstruction}} = \text{CD}(\mathbf{R}_m^g, \mathbf{P}_m^g) + \text{CD}(\mathbf{R}_m^l, \mathbf{P}_m^l), \quad (12)$$

where the losses from local block masking and global random masking are summed for computation. By utilizing generative modeling to enhance global contrastive learning like [13], our method effectively captures hierarchical structures and preserves semantic consistency, showcasing the advantages of hyperbolic geometry in point cloud representation learning.

3.3. Entailment loss

In order to enforce a partial order relationship between paired modals and further refining the hierarchical structure, we introduce cross-modal entailment loss in hyperbolic space. Fig. 4 illustrates the concept of entailment loss in a two-dimensional space.

Let $\mathbf{x}$ and $\mathbf{y}$ represent the embeddings of two modals. According to the definition of the conical region, the boundary of the cone is determined

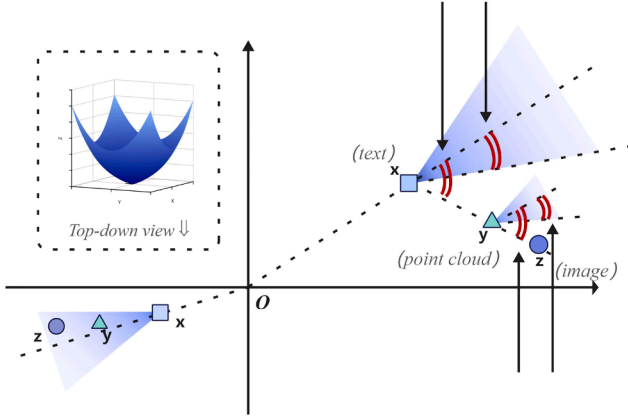


Fig. 4. The entailment loss in the Cartesian coordinate system. This loss pushes the point cloud embeddings into the conical region defined by the text embeddings, while simultaneously pushing the text embeddings into the conical region defined by the point cloud embeddings. Our goal is to ensure that the loss becomes zero if the image, point cloud, and text embeddings all lie within the conical region (third quadrant).

by the angle from the principal axis to the edge. These angles adhere to the geometric properties of hyperbolic space:

$$\sin(\psi(x)) = K \cdot \frac{1 - \|x\|^2}{\|x\|}, \quad (13)$$

where a constant $K = 0.1$ is used to set the boundary conditions near the origin. We define the entailment cone for each x , which narrows as we move farther from the origin. This cone is defined by the half-aperture:

$$\psi(x) = \arcsin \left(K \cdot \frac{1 - \|x\|^2}{\|x\|} \right), \quad (14)$$

noted that as x approaches the origin, $\psi(x)$ approaches its maximum value. As $\|x\| \rightarrow 1$, $\psi(x) \rightarrow 0$, and the conical region gradually contracts.

Our goal is to identify and penalize cases where the paired embedding y lies outside the entailment cone. To achieve this, we measure the exterior angle $\text{ext}(x, y) = \pi - \angle Oxy$, as shown in Fig. 4. Here, $\angle Oxy$ is the angle of the hyperbolic triangle formed by points O, x, y . Based on the cosine law for hyperbolic triangles, and considering the positions of x, y , the formula for the exterior angle can be expressed as:

$$\text{ext}(x, y) = \arccos \left(\frac{\langle x, y \rangle (1 + \|x\|^2) - \|x\|^2 (1 + \|y\|^2)}{\|x\| \cdot \|x - y\| \cdot \sqrt{1 + \|x\|^2 \|y\|^2 - 2\langle x, y \rangle}} \right). \quad (15)$$

Combining the half-aperture $\psi(x)$ and the exterior angle $\text{ext}(x, y)$, the entailment loss in the Poincaré ball is defined as:

$$\mathcal{L}_{\text{entail, Poincare}}(x, y) = \max(0, \text{ext}(x, y) - \psi(x)). \quad (16)$$

We aim to make the point cloud modal serves as a bridge. On one hand, the point cloud is constrained by textual abstraction, learning semantic concepts. On the other hand, the point cloud, by aligning with the image, learns specific geometric and visual details. The entailment should be composed of both textual-point cloud and point cloud-image relations.

$$\mathcal{L}_{\text{entail}} = \lambda_1 \mathcal{L}_{\text{text} \rightarrow \text{point}} + \lambda_2 \mathcal{L}_{\text{point} \rightarrow \text{image}}, \quad (17)$$

where λ_1 and λ_2 control the weights of the two components of the loss. Our overall loss function is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{reconstruction}} + \mathcal{L}_{\text{contrastive}} + \mathcal{L}_{\text{entail}}. \quad (18)$$

We adjust the back-propagation strategy for different encoders based on the characteristics of hyperbolic space. Specifically, for the 3D point cloud encoder, back-propagation is entirely disabled for the cross-attention mechanism throughout the process. For the 2D vision and text encoders, the parameters are frozen after 10 warm-up epochs to achieve a balance between efficiency and accuracy. Further details can be found in Table 13 in Ablation Study.

4. Experiments

4.1. Implementation details

Pretraining Dataset. Our model is pretrained on ShapeNet [12], which contains 55 categories and over 50,000 CAD models. For each point cloud in the dataset, we generate high-quality rendered images using MacOS Preview, along with category labels and manually crafted prompt templates. This method leverages the large-scale nature of RGB images and free-form language to significantly enrich the limited 3D data. Each point cloud in ShapeNet contains 2048 points with only x, y, z coordinates. All rendered images are resized to 224×224 pixels, with a patch size of 16 to fit the image encoder.

Model Architecture. We utilize a standard plain Transformer consisting of 12 HyperPoint-blocks with a dimension of 384, sharing weights across blocks and connected via cross-attention mechanisms. A lightweight PointNet [25] is employed for 3D patch embedding. We use Vision Transformer(ViT-B) [29] pretrained on ImageNet as the image encoder teacher and CLIP [4] as the text encoder teacher. Additionally, a two-layer MLP (384-128) is used as the projection head, resulting in a 128-dimensional hyperbolic space feature projection. For masked generative modeling, we use “FPS + KNN” to divide the input point cloud into 64 point patches, each containing 32 points. To demonstrate the effectiveness of hyperbolic contrastive learning, we adopt the simplest hybrid masking strategy in our experiments.

Experiment Details. During pretraining, each input sample pair consists of a 3D point cloud, its corresponding rendered image, and a textual category description. We only apply rotation augmentation to the 3D point clouds and no image augmentation is applied during pretraining. After the warm-up phase, the network parameters of the 2D image encoder and text encoder are frozen and excluded from gradient updates. Meanwhile, to prevent misleading gradient flows from the global hyperbolic contrastive task to the local reconstruction task, we apply the stop-gradient operation in cross-attention. We conduct pretraining over 300 epochs with a learning rate of 3×10^{-5} during 10 epochs warm-up and then increases to 5×10^{-4} . We use AdamW as the optimizer, a batch size of 64, and a drop path rate of 0.1. For contrastive learning, the curvature parameter is set to 0.1, temperature $\tau = 0.2$, and the hyperbolic embedding optimization $\lambda = 0.01$. All pretraining and downstream task experiments are conducted on NVIDIA GeForce RTX 4090s.

Notably, during pretraining, the parameters of the image and text encoder teachers remain learnable in the warm-up epochs. This decision is based on our observation that frozen pretraining parameters in Euclidean space are not entirely suitable for hyperbolic space. We also analyze the curvature parameter and other factors, with further details provided in Tables 10 and 12.

4.2. Transfer learning on downstream tasks

We evaluated the transferability and effectiveness of our HyperPoint framework on eight widely used downstream tasks in point cloud representation learning: (I) 3D Real-World Object Classification, (II) 3D Synthetic Object Classification, (III) Few-Shot Object Classification, (IV) 3D Zero-Shot Recognition, (V) 3D Part Segmentation (VI) Linear SVM Classification, (VII) 3D Scene Segmentation, (VIII) Robust classification.

3D Real-World Object Classification. ScanObjectNN [40] is one of the most challenging real-world datasets in the 3D domain, containing 15 categories and approximately 15,000 real-world objects. To ensure a fair comparison with other methods, we present the results both with and without employing a voting strategy. The results, summarized in Table 1, highlight several key findings. On three ScanObjectNN benchmark variants, our method improves performance by a substantial margin compared to three baseline models. Notably, in the OBJ_BG category, our model achieves an accuracy of **96.1 %** with voting, representing significant improvements over the cross-modal baseline (+1.78 %), and the MAE-based baseline (+6.08 %). On the PB_T50_RS category, our

Table 1

Classification results on the ScanObjectNN and ModelNet40 datasets. The dagger symbol [†] indicates that the model was reproduced under our setups. Overall accuracy (%) is reported for both without voting and with voting.

Method	#Params (M)	GFLOPS	ScanObjectNN			ModelNet40	
			OBJ_BG	OBJ_ONLY	PB_T50_RS	1k P	8k P
Supervised Learning Only							
PointNet [25]	3.5	0.5	73.3	79.2	68.0	89.2	90.8
PointNet + + [30]	1.5	1.7	82.3	84.3	77.9	90.7	91.9
DGCNN [28]	1.8	2.4	82.8	86.2	78.1	92.9	–
PointCNN [31]	0.6	–	86.1	85.5	78.5	92.2	–
SimpleView [32]	/	–	–	–	80.5	93.9	–
PCT [33]	2.88	2.3	–	–	–	93.2	–
PointMLP [34]	12.6	31.4	–	–	85.4	94.5	–
PointNeXt [26]	1.4	3.6	–	–	87.7	94.0	–
P2P-RN101 [35]	42.7	7.9	–	–	87.4	93.1	–
P2P-HorNet [35]	195.8	34.6	–	–	89.3	94.0	–
with Self-Supervised Representation Learning (MAE-based)							
Point-BERT [36]	22.1	4.8	87.43	88.12	83.07	93.2	93.8
MaskPoint [37]	22.1	4.8	89.30	88.10	84.30	93.8	–
Point-MAE [14]	22.1	4.8	90.02	88.29	85.18	93.8	94.0
Point-M2AE [17]	15.3	5.3	91.22	88.81	86.43	94.0	–
Improvement(baseline:PointMAE)	/	/	+ 6.08	+ 6.2	+ 5.45	+ 0.5	+ 0.1
with Self-Supervised Representation Learning (Hyperbolic)							
HyperIPC [22]	/	–	–	–	84.5	91.8	–
MERU(modified) [38]	/	–	–	–	–	93.6	94.3
Improvement(baseline:HyperIPC)	/	/	–	–	+ 6.13	+ 2.5	–
with Cross-Modal Self-Supervised Learning							
Joint-MAE [39]	/	–	90.94	88.86	86.07	94.0	–
I2P-MAE [18]	15.3	–	94.15	91.57	90.11	93.7	–
ACT [6]	22.1	4.8	93.29	91.91	88.21	93.7	94.0
RECON† w/o vot. [13]	44.3	5.3	93.63	93.29	89.69	93.5	93.6
RECON† w/ vot. [13]	44.3	5.3	94.32	93.63	90.22	93.6	93.7
HyperPoint w/o vot.	50.7	5.6	95.52	94.32	90.35	94.2	94.0
HyperPoint w/ vot.	50.7	5.6	96.10	94.49	90.63	94.3	94.1
Improvement(baseline:RECON)	/	/	+ 1.78	+ 0.86	+ 0.41	+ 0.7	+ 0.4

model achieves +6.13% improvement compared to hyperbolic-based methods.

3D Synthetic Object Classification. The ModelNet [41] dataset, introduced by Wu et al. in 2015, is a foundational benchmark for synthetic 3D object recognition. It consists of 12,000 meshed 3D CAD models, categorized into 40 classes in the ModelNet40 configuration. Our evaluation focuses on two key tasks: fine-tuning for object classification and few-shot learning, both conducted on ModelNet40 to evaluate the effectiveness of our proposed methods. During fine-tuning, we follow the procedures outlined in PointNet, applying scale and translate operations for data augmentation. Table 1 demonstrates that our HyperPoint model achieves an accuracy of 94.2% without voting and **94.3%** with voting. This performance surpasses all baseline methods by a significant margin, with an overall improvement of +3.7%. These results highlight the strong generalization and recognition capabilities of our model within the ModelNet40 dataset.

Few-Shot Object Classification. We investigate the Few-Shot Learning (FSL) paradigm through experiments conducted on the ModelNet40 [41] dataset. Our method involves selecting a subset of classes at random and using a minimal number of samples from each class, thereby constructing a constrained training dataset to evaluate the model's generalization capacity. The experimental framework aligns with a decade of FSL challenges, and we present a comprehensive analysis, including both mean accuracy and standard deviation. Table 2 summarizes the FSL results on ModelNet40 under experimental conditions where the number of classes (N) is restricted to 5, 10 and the number of samples per class (M) is limited to 10, 20. The results show that, under the same parameter constraints, our HyperPoint model achieves new state-of-the-art performance, surpassing existing methods by +0.4%, +0.2%, +0.8%, and +0.2% on four downstream tasks, respectively.

Additionally, our HyperPoint model demonstrates high consistency and reliability, achieving the lowest standard deviations of ± 1.9 for 10 shots and ± 0.9 for 20 shots in the 5-way task. Following [43], we run 600 random episodes and record accuracy. We then apply 1000 bootstrap samples to obtain the mean distribution and report 95% confidence intervals. Table 2 shows that even the widest CI (10-way 10-shot) has a half-width of only 0.9%, indicating that 600 episodes are sufficient to stabilize the estimate. This confirms that our method yields more stable results than the baseline.

3D Zero-Shot Recognition. Like MERU [20], our model aligns the feature spaces of textual and other modals in the hyperbolic space. Therefore, our model demonstrates a more efficient capability to represent data compared to Euclidean space. As shown in Table 3, when freezing the backbone model and fine-tuning only the embedding dimensions for the zero-shot task, the performance of our model declines more slowly across all embedding dimensions. This reflects that the hyperbolic space is more robust to low-dimensional embeddings, retaining more hierarchical and semantic information in lower dimensions. Even with a significant reduction in dimensions, our model still outperforms the baseline, indicating that the hyperbolic space can utilize the embedding space more efficiently.

3D Part Segmentation. To evaluate geometric understanding at the object scale, we conducted part segmentation experiments on the ShapeNetPart [49] dataset. Specifically, we fused cross-modal features in hyperbolic space into global representations and applied the same segmentation head as used in PointMAE [14]. As shown in Table 4, our model outperforms the baseline by +0.2% in Cls.mIoU and +0.4% in Inst.mIoU. This improvement can be attributed to the superior geometric representation capabilities of hyperbolic space, enabling HyperPoint to achieve enhanced shaping understanding.

Table 2

Few-shot learning results on the ModelNet40 dataset. The table reports the average classification accuracy (%) along with the standard deviation (%) across four downstream experiments.

Method	5-way		10-way	
	10-shot	20-shot	10-shot	20-shot
<i>Supervised Learning Only</i>				
PointNet [25]	52.0 $\pm$ 3.8	57.8 $\pm$ 4.9	46.6 $\pm$ 4.3	35.2 $\pm$ 4.8
DGCNN [28]	31.6 $\pm$ 2.8	40.8 $\pm$ 4.6	19.9 $\pm$ 2.1	16.9 $\pm$ 1.5
CrossPoint [5]	92.5 $\pm$ 3.0	94.9 $\pm$ 2.1	83.6 $\pm$ 5.3	87.9 $\pm$ 4.2
<i>Single-Modal Self-Supervised Learning</i>				
OcCo [42]	94.0 $\pm$ 3.6	95.9 $\pm$ 2.3	89.4 $\pm$ 5.1	92.4 $\pm$ 4.6
Point-BERT [36]	94.6 $\pm$ 3.1	96.3 $\pm$ 2.7	91.0 $\pm$ 5.4	92.7 $\pm$ 5.1
MaskPoint [37]	95.0 $\pm$ 3.7	97.2 $\pm$ 1.7	91.4 $\pm$ 4.0	93.4 $\pm$ 3.5
Point-MAE [14]	96.3 $\pm$ 2.5	97.8 $\pm$ 1.8	92.6 $\pm$ 4.1	95.0 $\pm$ 3.0
Point-M2AE [17]	96.8 $\pm$ 1.8	98.3 $\pm$ 1.4	92.3 $\pm$ 4.5	95.0 $\pm$ 3.0
Point-FEMAE [27]	97.2 $\pm$ 1.9	98.6 $\pm$ 1.3	94.0 $\pm$ 3.3	95.8 $\pm$ 2.8
Improvement(baseline:PointMAE)	+1.1	+1.1	+1.5	+0.9
<i>Cross-Modal Self-Supervised Learning</i>				
MERU(modified) [38]	97.4	99.1	93.5	95.9
ACT [6]	96.8 $\pm$ 2.3	98.0 $\pm$ 1.4	93.3 $\pm$ 4.0	95.6 $\pm$ 2.8
Joint-MAE [39]	96.7 $\pm$ 2.2	97.9 $\pm$ 1.8	92.6 $\pm$ 3.7	95.1 $\pm$ 2.6
I2P-MAE [18]	97.0 $\pm$ 1.8	98.3 $\pm$ 1.3	92.6 $\pm$ 5.0	95.5 $\pm$ 3.0
RECON [†] [13]	97.0 $\pm$ 1.9	98.7 $\pm$ 1.2	93.3 $\pm$ 3.9	95.7 $\pm$ 3.0
HyperPoint	97.4 $\pm$ 1.9	98.9 $\pm$ 0.9	94.1 $\pm$ 4.4	95.9 $\pm$ 2.2
Bootstrap 95% CI	[96.9, 98.0]	[98.7, 99.2]	[93.2, 95.0]	[95.3, 96.5]
Improvement(baseline:RECON)	+0.4	+0.2	+0.8	+0.2

Table 3

Zero-shot 3D object classification on the ModelNet40 dataset across varying embedding dimensions. The Top-1 accuracy (%) is reported. HyperPoint outperforms the baseline at lower embedding dimensions.

Method	Embedding dimensions				
	512	256	128	96	64
PointCLIP [44]	–	20.2	–	–	–
CLIP2Point [45]	–	49.4	–	–	–
PointNet++ + ULIP [30,46]	–	55.7	53.7	52.2	48.0
PointMLP + ULIP [34,46]	–	61.5	59.1	55.8	53.7
PointBERT + ULIP [36,46]	–	60.4	59.0	58.4	55.7
RECON [13]	64.6	64.4	64.0	63.2	59.4
HyperPoint	64.8	64.4	64.2	63.6	61.8

Table 4

Part segmentation on ShapeNetPart dataset. The mIoU(%) overall classes (Cls.) and the mIoU(%) over all instances (Inst.) are reported.

Methods	Cls. mIoU (%)	Inst. mIoU (%)
PointNet [25]	80.4	83.7
PointNet++ + [30]	81.9	85.1
DGCNN [28]	82.3	85.2
PointMLP [34]	84.6	86.1
Transformer [47]	83.6	85.2
PointContrast [16]	–	85.1
CrossPoint [5]	–	85.5
Point-BERT [36]	84.1	85.6
ACT [6]	84.7	86.1
Point-MAE [14]	84.4	86.1
EDGCNet [48]	84.4	86.7
RECON [13]	84.8	86.4
HyperPoint	85.0	86.8

Linear SVM Classification: Linear SVM serves as a metric to assess the discriminative power of pretrained features. As detailed in Table 5, our Hyperpoint model outperforms Point-BERT [36] by a clear margin of +5.8%, which employs plain Transformers. For ScanObjectNN, Hyperpoint achieves a remarkable lead over PointMAE [14], with an improvement of +8.8%. When compared to methods utilizing

Table 5

Linear SVM evaluation results on ModelNet40 and ScanObjectNN (OBJ_BG).

Method	OBJ_BG	ModelNet40
OcCo [42]	69.5	89.2
Point-BERT [36]	–	87.4
Point-MAE [14]	77.7	91.0
CrossPoint [5]	75.6	91.2
PointM2AE [17]	84.1	92.9
Joint-MAE [39]	–	92.4
RECON [13]	–	93.4
I2P-MAE [18]	87.1	93.4
PTM [50]	87.9	93.8
Hyperpoint	86.5	93.2
Improvement(baseline:PointMAE)	+8.8	+2.2

hierarchical Transformers, Hyperpoint surpasses PointM2AE [17] by +2.4%, +0.3% respectively. These results underscore the discriminative capabilities of our model, highlighting its representation capability over hyperbolic space.

3D Scene Segmentation: Scene segmentation poses a significant challenge, particularly in large-scale 3D environments, due to the necessity for models to understand complex contextual semantics and detailed geometric interactions. We conduct 3D scene segmentation on the S3DIS [51] dataset, which is a large-scale indoor point cloud dataset created by the Stanford University, covering abundant scenes within six different buildings. As detailed in Table 6, our HyperPoint model demonstrates superior performance, achieving a 1.2% increase in mean accuracy (mAcc) and a 0.4% improvement in mean intersection over union (mIoU) compared to PointMAE [14].

ModelNet40-C Corruptions. We use ModelNet40-C [52] to evaluate robustness against natural corruptions. It covers three LiDAR-like corruptions: density variation, noise, and transformation. We take the model trained on clean ModelNet40 and test it on ModelNet40-C without retraining. Besides adversarial training, we include PointMixup [53], PointCAT [54] as baselines. Table 7 summarizes the results. Without extra augmentation or adversarial training, HyperPoint matches PointMixup [53] in robust accuracy, confirming its strong performance under natural corruptions.

Table 6
Semantic segmentation results on S3DIS area 5.

Method	Semantic Seg.	
	mAcc	mIoU
PointNet [25]	49.0	41.1
PointNet++ [30]	67.1	53.5
Transformer [47]	68.6	60.0
PointBERT [36]	70.0	60.6
MaskPoint [37]	70.0	60.3
PointMAE [14]	69.9	60.8
ACT [6]	71.1	61.2
HyperPoint	71.1	61.2
Improvement(baseline:PointMAE)	+1.2	+0.4

Table 7
Experimental results of robust accuracy (%) against common corruptions on ModelNet40-C.

Robust Setting	Split Acc (%)			Robust Accuracy
	Density	Noise	Trans.	
No-Defense	71.7	67.3	76.2	71.7
PointMixup [53]	71.7	71.1	81.0	74.6
BN [55]	73.2	70.4	78.8	74.1
AT [56]	73.3	69.8	80.5	74.7
PointCAT [54]	69.6	76.6	77.8	74.8
HyperPoint	72.2	70.4	77.0	74.6

Table 8
Evaluation of hierarchical information distortion and classification accuracy using RECON [13] as $\mathcal{L}_{\text{Flat}}$. All metrics are reported as mean over 3 seeds.

Dataset	Method	Distortion of Hierarchy	Classification Accuracy
		$\delta_{\text{rel}}(\downarrow)$	Fine ($\uparrow$)
ShapeNet	Flat	0.203	N/A
	HyperPoint	0.058	N/A
ModelNet40	Flat	0.214	93.5
	HyperPoint	0.061	94.2
ScanObjectNN	Flat	0.237	93.63
	HyperPoint	0.073	95.52

4.3. Ablations and analysis

Tree-likeness. To quantify how tree-like the learned representations are, we report Gromov hyperbolicity δ_{rel} [9,11] in Table 8. Smaller δ_{rel} means stronger tree structure, $\delta_{\text{rel}} = 0$ for a perfect tree metric. HyperPoint cuts hierarchical distortion by approximately 70 % compared with Flat and boosts classification accuracy on every dataset. We demonstrate that accurate embedding of hierarchical relations enhances classification accuracy, producing features that adhere to the label hierarchy and exhibiting lower distortion.

Cross-Modal Retrieval. As stated in Section 3.3, HyperPoint readily supports cross-modal downstream tasks. Fig. 5 qualitatively demonstrates its potential for CAD image to point cloud retrieval. We adopt the pretrained HyperPoint as a frozen 3D encoder. We conduct a small-scale study using CAD images from RECON [13] to retrieve 3D point clouds from roughly 52 k ShapeNet samples. We show the top-5 retrieved models for query images from the airplane and table categories. The results confirm that the pretrained model learns meaningful cross-modal features. Surprisingly, the top-5 models returned by HyperPoint are visually closer to the query image than those retrieved in Euclidean space. For example, when a civil airplane image is used, HyperPoint captures fine-grained details such as four-engine propellers and low-mounted wings, whereas the last three Euclidean matches miss these cues.

Align Representations. As described in Eq. (7), HyperPoint by default aligns the 3D representation with both the text and image

Table 9

Analysis of aligning three vs. two modalities on three benchmarks. Results show that aligning representations of three modalities always produces better results than two modalities.

Model	Modality aligned	ModelNet40	ScanObjectNN	ShapeNetPart
HyperPoint	P + T	92.8	93.29	84.4
	P + I	93.4	94.15	84.4
	P + I + T	94.2	95.52	85.0

Table 10

Ablation study on curvatures. The table presents the overall accuracy (%) without voting on the ScanObjectNN OBJ_BG dataset.

C	0.01	0.1	0.3	0.5	1.0
Accuracy	94.32	95.52	91.91	91.57	90.11

representations during pre-training. We wonder to what extent HyperPoint will still work if we align the 3D representation to only the text feature or image feature alone. We conduct an ablation study for HyperPoint by aligning two rather than three modalities in standard settings. Results are shown in Table 9 for three benchmarks. Aligning the 3D modality with both text and image modalities consistently achieves the best performance compared to aligning with either image or text modality in every scenario with each baseline.

Entailment Loss. We employ only the contrastive loss and reconstruction loss for training, which effectively corresponds to setting $\lambda = 0$. From Table 11, the removal of entailment loss is shown to have a significant impact on the overall performance of HyperPoint. This demonstrates that the entailment loss is essential for achieving superior quantitative performance.

Hyperparameters. We present an ablation study in Fig. 6, exploring the effects of masking ratio, decoder depth, the number of warm-up epochs during pretraining, and loss multiplier λ . The results indicate that the optimal masking ratio and decoder depth are consistent with the parameters reported in PointMAE, demonstrating the strong transferability of our chosen model. Furthermore, the results highlight that the selected number of warm-up epochs strikes a fine balance, indicating a potential trade-off between model complexity and performance. We set $\lambda_1 = \lambda_2 = 0.2$, larger values of λ impose stronger regularization on the hyperbolic contrastive loss, which negatively impacts performance.

Curvatures. Table 10 presents the performance of our model on the ScanObjectNN-OBJ_BG under varying curvature values. The results indicate that the model exhibits strong robustness when the curvature is small, whereas larger curvature values lead to performance degradation. This suggests that when the curvature is sufficiently small and approaches zero, the hyperbolic space achieves a larger effective radius, providing greater stability for the data and enhancing model performance.

Pretraining Targets. To further analyze the significance of each module and validate the effectiveness of our reconstruction-guided hyperbolic contrastive learning, we performed ablation experiments on the model's main components. Table 12 reveals that performing generative-guided contrastive learning in Euclidean space instead of hyperbolic space significantly deteriorates the model's performance, underscoring the effectiveness of our hyperbolic projection. Removing the point cloud self-contrastive learning component results in decreased accuracy, which can be attributed to the exponential nature of hyperbolic space that overcomes the limitations of Euclidean space for point cloud data. This property enables the contrastive learning task to converge more effectively to an optimal solution. Excluding the generative-guided learning method leads to a significant drop in accuracy, validating that the generative modeling task enhances the generalization capability of point cloud representations.

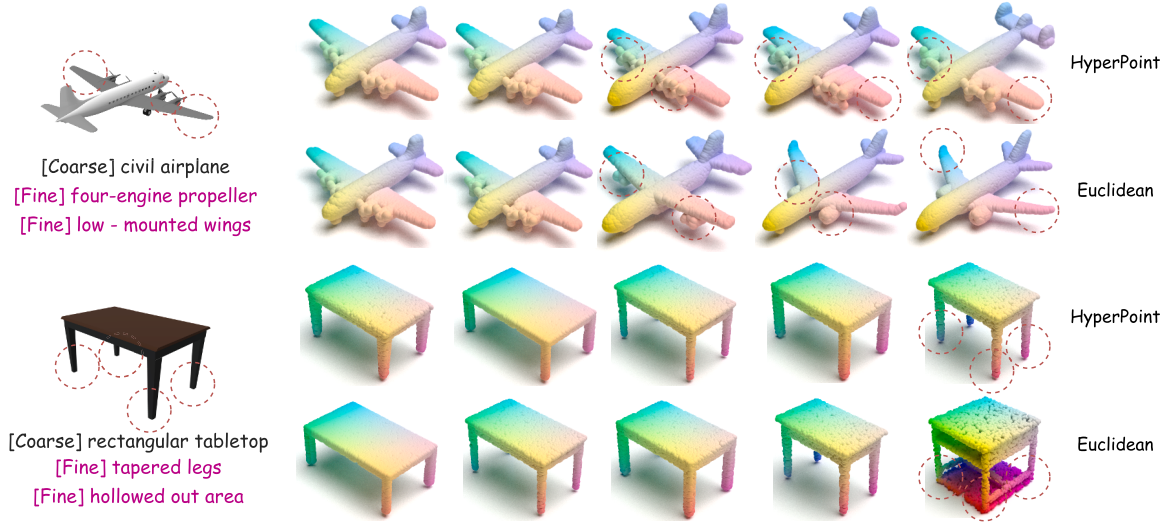


Fig. 5. Qualitative results of CAD image to point cloud retrieval. Query images are from RECON and point clouds are from ShapeNet. We show the top-5 retrieved point cloud models, ranked in order. The results demonstrate the retrieval capability of our model.

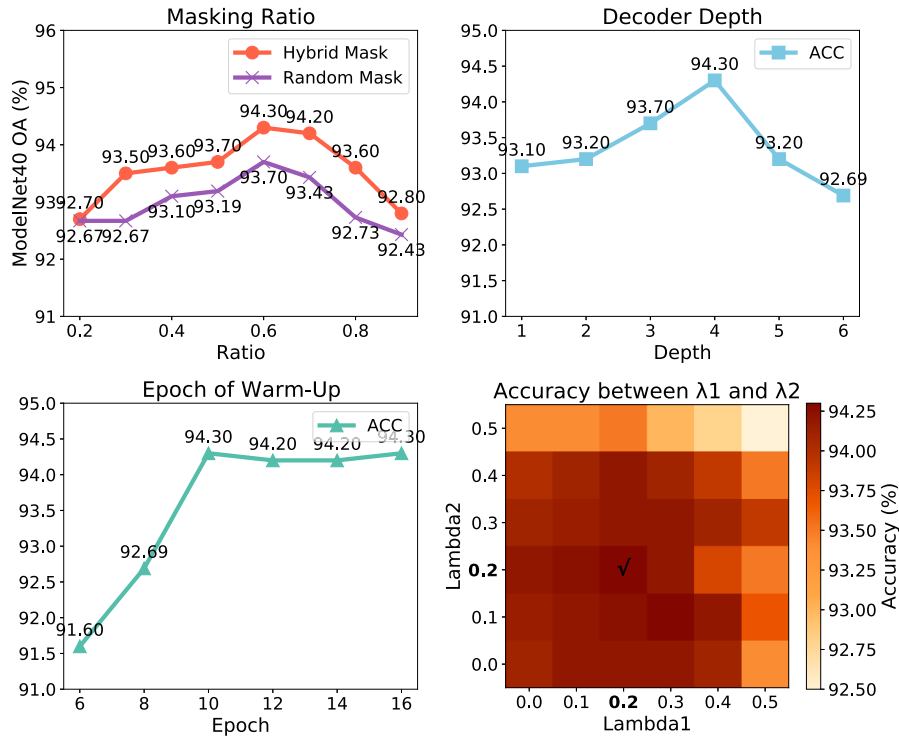


Fig. 6. Ablation study on the masking ratio, decoder depth, the selection of warm-up epochs during pretraining, and loss multiplier λ . The masking ratio includes two strategies: the hybrid masking strategy (ours) and the global random masking strategy (PointMAE). Decoder depth refers to the generative mask modeling decoder. The parameters of the text encoder and image encoder are frozen after warm-up epochs. $\lambda > 0$ is necessary to induce partial order in hyperbolic space.

Table 11

Ablation study with and without entailment loss across downstream tasks w/ voting strategy. Entailment loss is pivotal for enhancing the interpret-ability of the hyperbolic structure.

Method vot.	ScanObjectNN			ModelNet40	
	OBJ_BG	OBJ_ONLY	PB_T50_RS	1k P	8k P
HyperPoint w/ entailment loss	96.10	94.49	90.63	94.3	94.1
HyperPoint w/o entailment loss	94.66 ↓ 1.44	93.28 ↓ 1.21	89.69 ↓ 0.94	94.1 ↓ 0.2	94.0 ↓ 0.1

Table 12

Ablation study on pretraining targets. The table reports the overall accuracy (%) without voting on the ScanObjectNN OBJ_BG dataset.

Hyperbolic	MAE	Contrastive		ScanObjectNN OBJ_BG	MN40 1k P
		text&img	self		
×	×	×	✓	85.57	89.2
×	×	✓	✓	86.02	92.2
×	✓	✓	✓	93.63	93.1
✓	✓	✓	×	93.29	93.8
✓	×	✓	✓	90.18	93.2
✓	✓	×	✓	93.63	93.6
✓	✓	✓	✓	95.52	94.2

Table 13

Ablation study on back-propagation operations and backbone. The table reports the overall accuracy (%) without voting on the ScanObjectNN OBJ_BG dataset.

Backbone	Back-propagation		Accuracy
	3D Encoder	2D & Text Encoder	
ViT-B & CLIP	✓	✓	90.02
ViT-B & CLIP	×	✓	95.52
ViT-B & CLIP	×	×	93.29
Resnet34 & CLIP	✓	✓	87.13
Resnet34 & CLIP	×	✓	94.15
Resnet34 & CLIP	×	×	93.29

Table 14

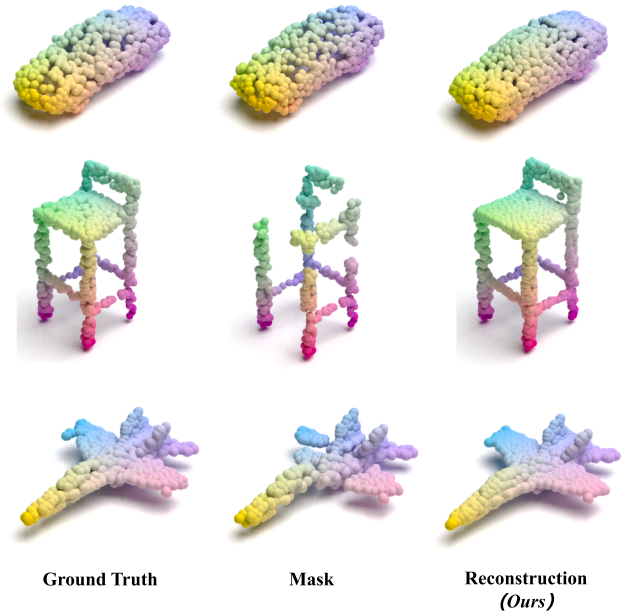
Ablation study of data augmentations (DA) during fine-tuning. We report the fine-tuning overall accuracy (%) w/ voting of pretrained models.

Augmentation Strategy	ScanObjectNN	ModelNet40
None	87.65	93.6
Scale & Translate	87.22	94.1
Jitter	89.92	93.1
Rotation	90.63	92.5
Dropout	87.60	93.7
Horizontal Flip	87.47	93.8

Encoder with Back-Propagation Operation. In addition to the aforementioned experiments, we explored strategies for identifying better 2D and text encoder teachers to further enhance model performance. The results, summarized in Table 13, reveal that using a VisionTransformer (ViT-B) as the 2D encoder teacher outperforms architectures based on ResNet. Additionally, we observe that employing back-propagation operations on 3D encoder directly reduces model performance. We argue that this decline is due to the fundamental difference between the metrics used in hyperbolic contrastive learning tasks and those in Euclidean generative modeling tasks. Back-propagation introduces unavoidable noisy gradients into the generative modeling task, thereby adversely affecting its performance. Similarly, directly performing exponential mapping without adjusting the parameters of CLIP and ViT is undesirable. It is necessary to allow 10 epochs for the parameters to adjust their weights. Although this increases the model's complexity and parameter count during the warm-up phase, it is a crucial step for ensuring the effectiveness of the transition.

Data Augmentation We investigate the effect of various data augmentation (DA) techniques during the fine-tuning of HyperPoint pretrained models on different downstream tasks. The results, as presented in Table 14, indicate that *Rotation* achieves the highest performance improvement on ScanObjectNN, while *Scale&Translate* delivers the best results on ModelNet40. Consequently, in the absence of specific requirements, we adopt *Rotation* as the default augmentation for ScanObjectNN and *Scale&Translate* for ModelNet40.

Reconstruction Visualization. Fig. 7 illustrates the results of random masking within our HyperPoint method, alongside the outcomes of

**Fig. 7.** Visualization of random masking and reconstruction of HyperPoint.

the subsequent reconstruction process. The findings demonstrate that the reconstructions produced by HyperPoint closely approximate the actual data, to the extent that they are virtually indistinguishable. This observation underscores the efficacy of HyperPoint in capturing detailed three-dimensional representations, thereby achieving superior modeling outcomes.

5. Conclusion

In this work, we introduce HyperPoint, the first multi-modal 3D foundation model in hyperbolic space. HyperPoint leverages the unique properties of hyperbolic space to enhance feature extraction and representation. Our method integrates generative and contrastive learning within a hyperbolic geometry framework, allowing for the efficient utilization of cross-modal data and the preservation of hierarchical semantic information. We note that HyperPoint may degrade on complex natural scenes where clean semantic trees are unavailable. As a result, we will explore leveraging hyperbolic hierarchies in larger-scale data in the future. We hope this work opens a new avenue for self-supervised 3D representation learning in hyperbolic space.

CRedit authorship contribution statement

Yiding Sun: Writing – original draft, Conceptualization; **Haozhe Cheng:** Writing – review & editing; **Chaoyi Lu:** Conceptualization; **Zhengqiao Li:** Data curation; **Minghong Wu:** Formal analysis; **Huimin Lu:** Supervision; **Jihua Zhu:** Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.patcog.2025.112800](https://doi.org/10.1016/j.patcog.2025.112800).

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